

DS 313: Statistical Foundations of Data Science

Course Code	DS 313
Course Name	Statistical Foundations of Data Science
L-T-P-C	3-1-0-4
Prerequisites	IC110-Engineering Mathematics, IC252-Data Science 2
Intended for	Discipline core for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines; From 2022 UG Batches
Approval	55th BoA (May 2024)

Course Contents

- **Module I:** Sample space, Sigma field, axiomatic definition of probability, conditional probability and independence, Bayes Rule. (2 lectures)
- **Review:** Random variables – discrete and continuous, probability mass function, probability density function, some standard (important) pdfs, independence, expectation, variance, conditional distribution, conditional expectation, covariance and correlation, Functions of random variables. (7 lectures)
- **Module III:** Probability generating function, moment generating function and characteristic functions – properties and applications. (3 lectures)
- **Module IV:** Convergence of random variables – basic results, inequalities (Markov and Chebyshev), law of large numbers (weak and strong), central limit theorem. (5 lectures)
- **Module V:** Random vectors and covariance and correlation matrix, Random processes – stationarity, WSS, Autocorrelation, cross correlation, power spectral density, Ergodicity. Wiener processes, Markov processes, Poisson Process. (8 lectures)
- **Sampling methods:** Inverse transforms sampling, Rejection sampling, adaptive rejection sampling, importance sampling, Markov chains and MCMC (8 lectures)
- **Graphical models:** ML and MAP estimation, directed and undirected models, Bayesian networks, CRF, Learning and Inference method (ML, MAP, Sampling) (9 lectures)

Text Books

1. Grimmett, Geoffrey, and David Stirzaker, **Probability and random processes**, Oxford university press, 2001.
2. Bishop, Christopher M., **Pattern recognition and machine learning**, Springer, 2006.

References

1. Ross, Sheldon., **A first course in probability**, Pearson, 2014.
2. Stark, Henry, and John William Woods, **Probability, statistics, and random processes for engineers**, Pearson, 2012.
3. Papoulis, Athanasios, and S. Unnikrishna Pillai, **Probability, random variables, and stochastic processes**, Tata McGraw-Hill Education, 2002.